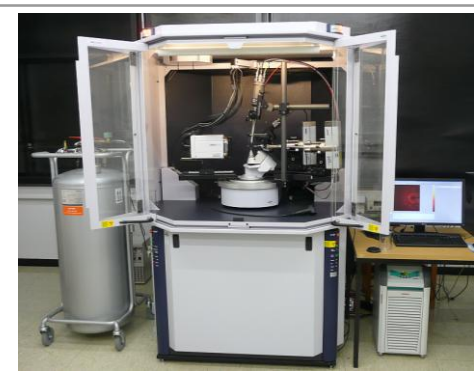
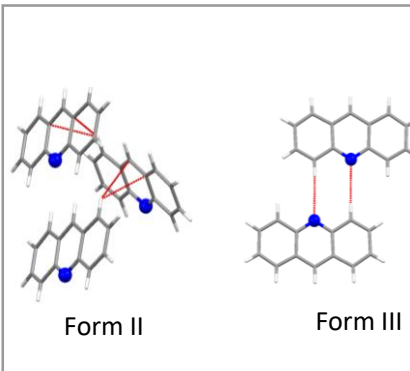
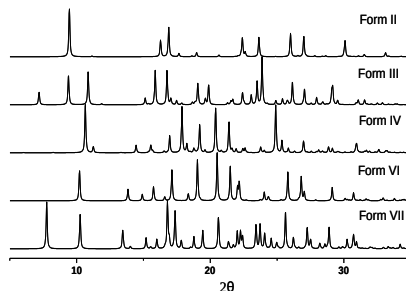
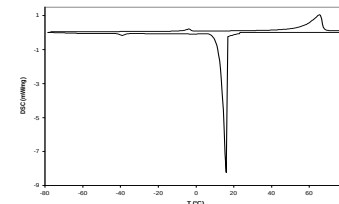
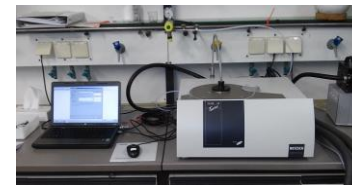
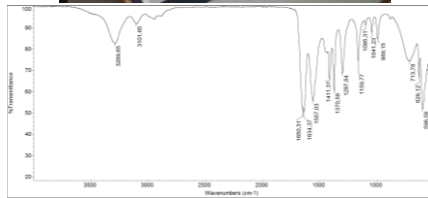
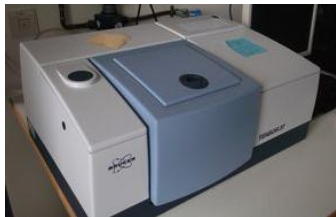


Methods of Characterization

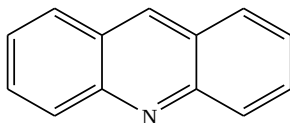
➤ Crystals, Polymorphs, Co-crystals

- Spectroscopy – **IR** and **Raman**
- Thermal methods – **DSC** and **TGA**
- Powder X-Ray Diffraction – **PXRD**
- Single Crystal X-Ray Diffraction – **SC-XRD**

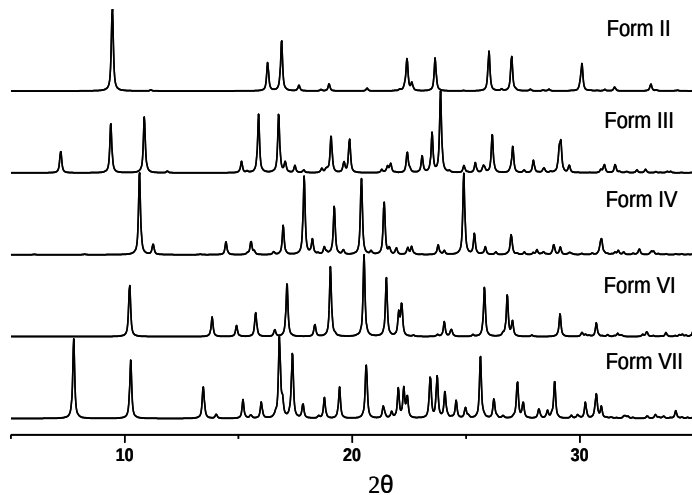


Powder X-Ray diffraction

➤ for polymorphism check



acridine



Different PXRD pattern for different forms

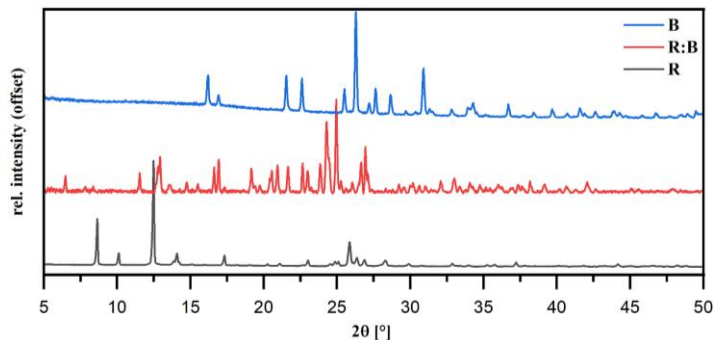
- Identification of **new polymorphs**
- Identification of **mixtures**
- Solid-state purity & cristallinity

HOW?

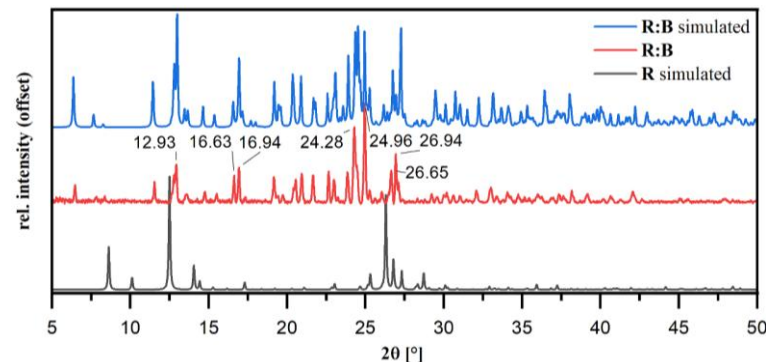
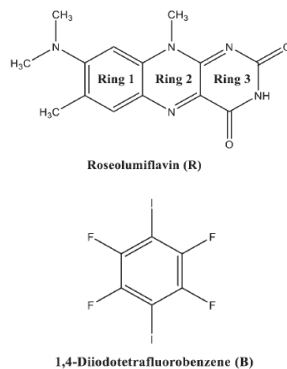
- comaparison of overall pattern
- representative first 2-3 reflections at low 2theta angle & most intense reflections
- check background (high or low)

Powder X-Ray diffraction

➤ for co-crystal identification



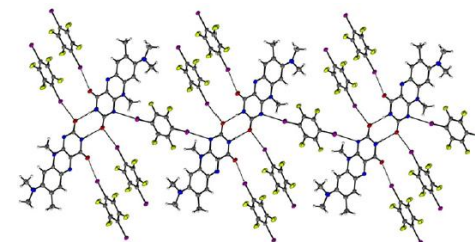
Experimental powder pattern of **R:B**
compared to experimental patterns of **R** and **B**



Experimental patterns of **R:B** compared to
simulated **R** and **R:B** from single crystal

Different PXRD pattern for new MCCs

- Identification of **new co-crystals** vs. physical mixtures of starting materials
- Phase purity and crystallinity via comparison with simulated powder pattern from single-crystal data



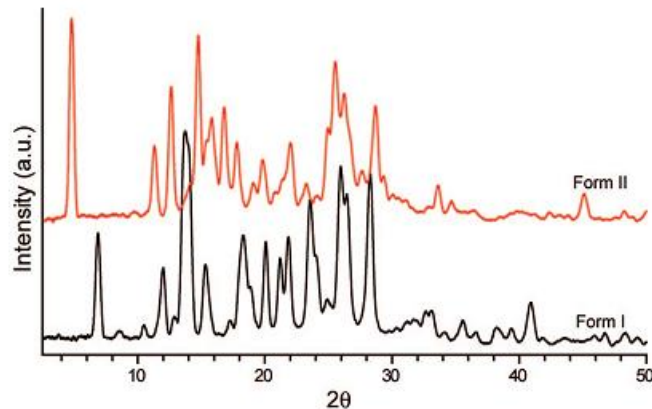


Figure 3. Powder X-ray diffraction patterns of CBZ-SAC I and CBZ-SAC II.

Differences in crystal **morphology**:

- plates (Form I)
- needles (Form II)

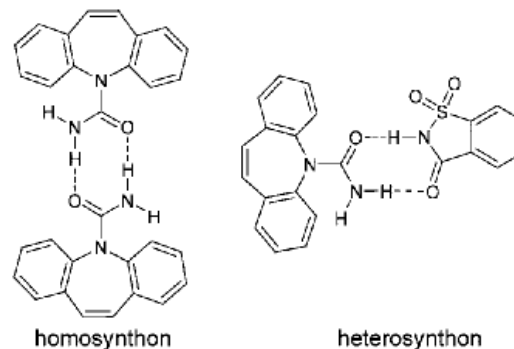
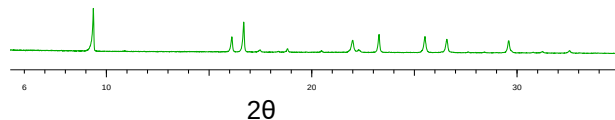
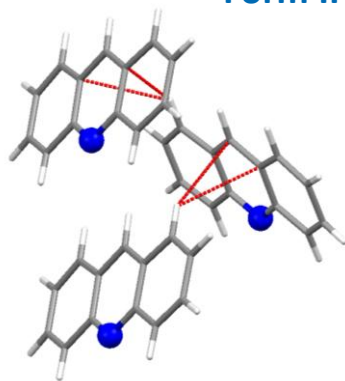


Figure 5. Representation of the homosynthon between two CBZ molecules in CBZ-SAC I and the heterosynthon between a CBZ and SAC molecule in CBZ-SAC II.

Acridine in acetone

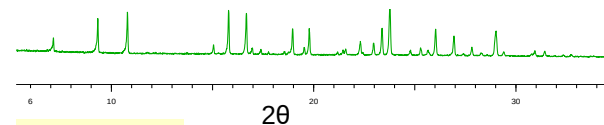


Form II

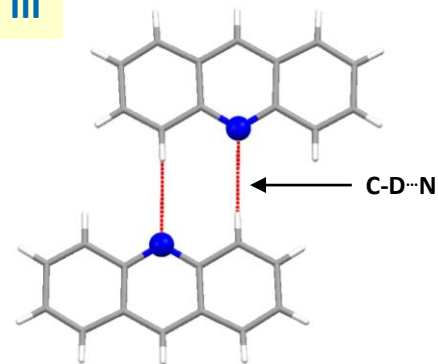


the most stable polymorph

d_9 -Acridine in acetone



Form III

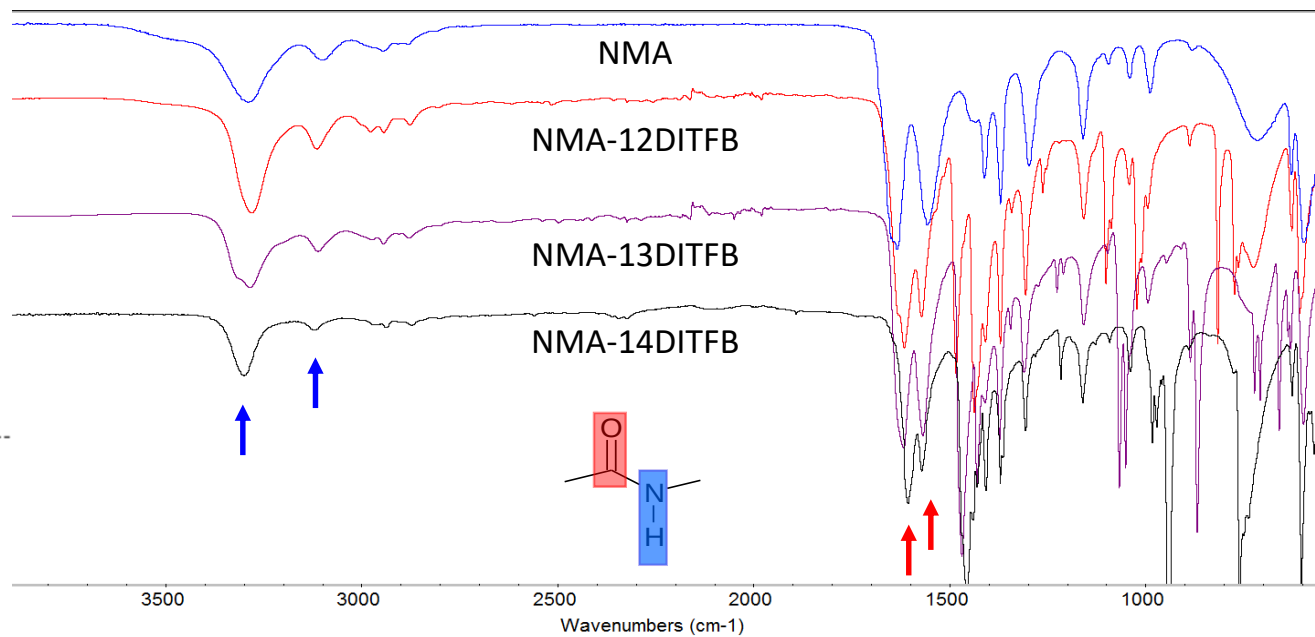
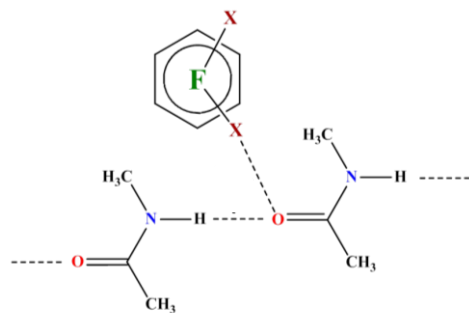


stable at room temperature

⇒ Stabilization of the less stable form III by N-D $\cdots$ C intermolecular interactions

IR-Spectroscopy for MCC

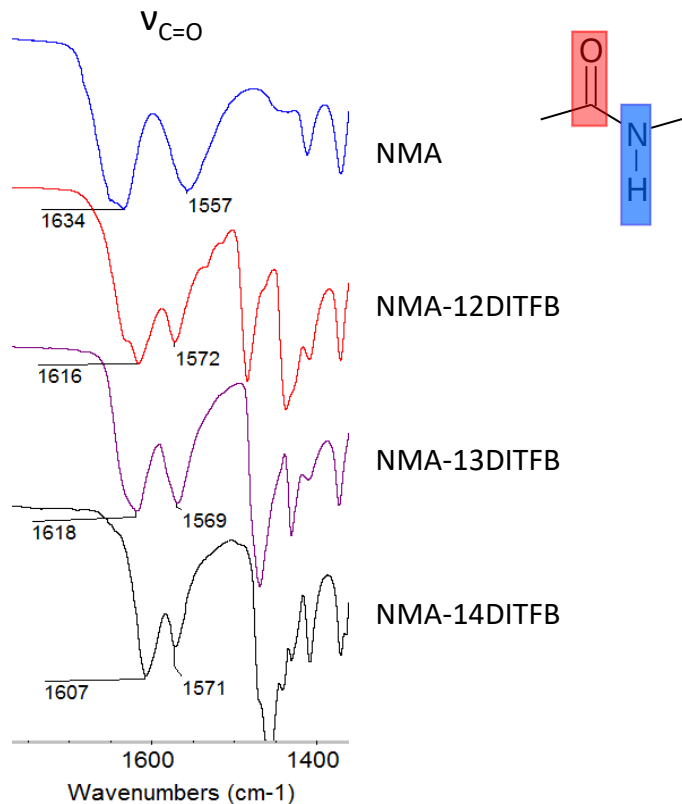
➤ NMA / diiodotetrafluorobenzene co-crystals



Shifts of the N-H and C=O vibration bands

IR shifts

NMA-DHaITFB-Systems



	$\nu_{\text{C=O}}$ (cm ⁻¹)	$\nu_{\text{N-H}}$ (cm ⁻¹)
NMA	1634	1557
NMA-12DITFB	1616	1572
NMA-13DITFB	1617	1569
NMA-14DITFB	1607	1571

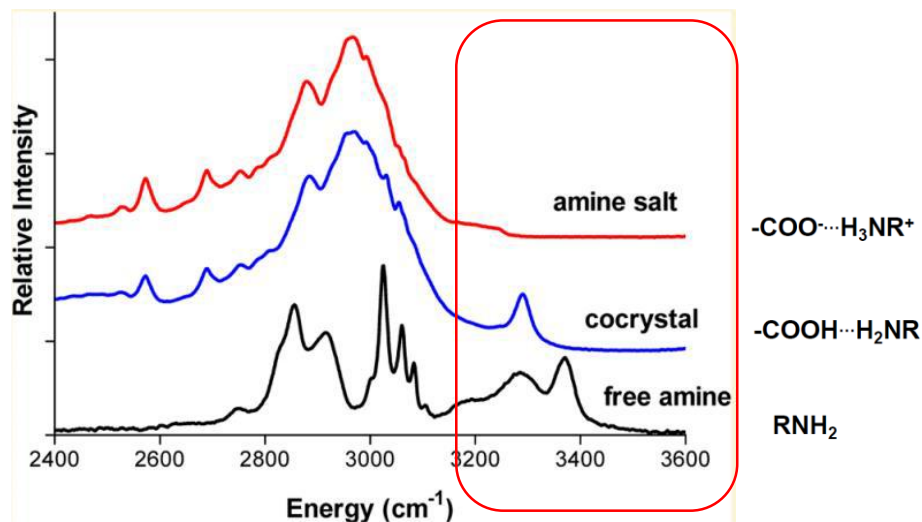
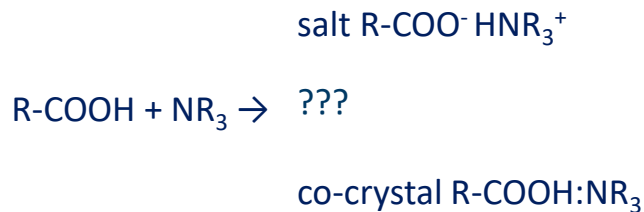
- blue shift of the N-H band
- red shift of the C=O band

General pattern is unchanged

- sum of IR vibration of both components
- **changes in interaction modes and strengths**
 - shifts of the band position (> 3 cm⁻¹)
 - possible alteration of the band form

IR for MC-systems with ionic character

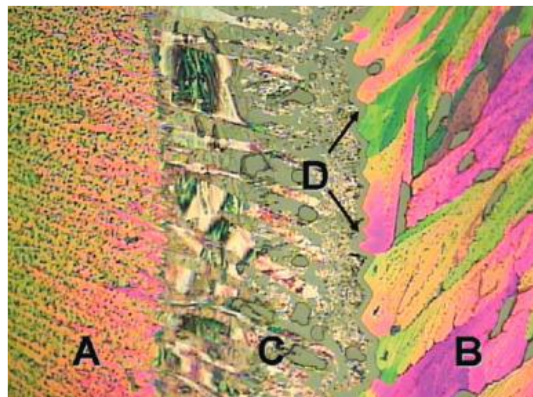
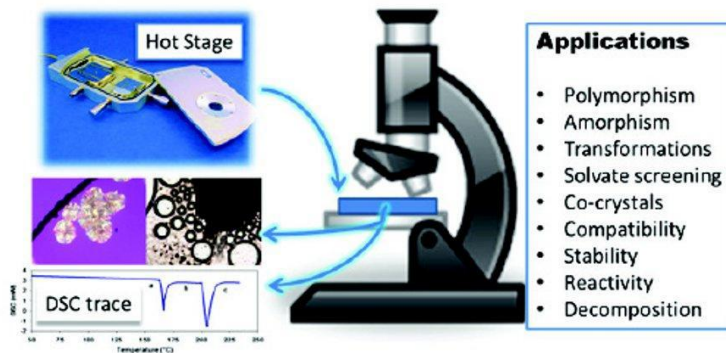
- distinguish between “salt” vs “co-crystal” formation



What about polymorphs?

How can we thermally prove polymorphs and MCCs?

Hot Stage Microscopy (HSM) in real-time

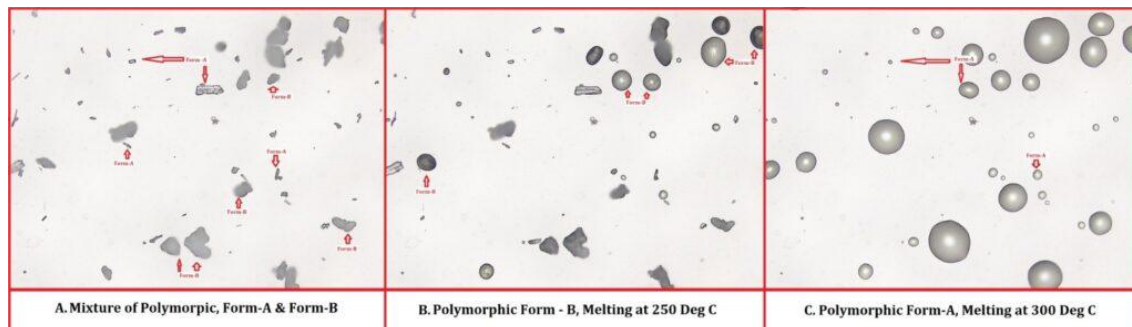


Contact area at 205°C

- A and B: two components
- C: **co-crystal**
- D: eutectic C:B starts to melt

Identification of **polymorphic forms**

- A: mixture of 2 forms
- B and C: thermal transitions
→ melting at different temperatures



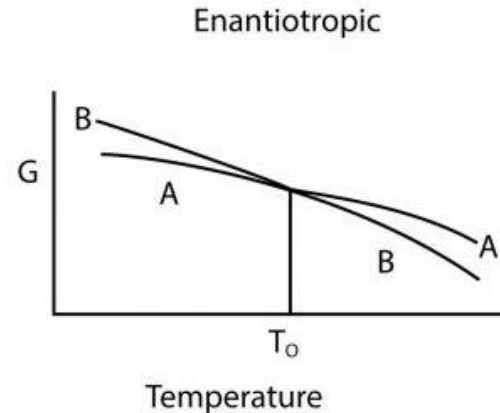
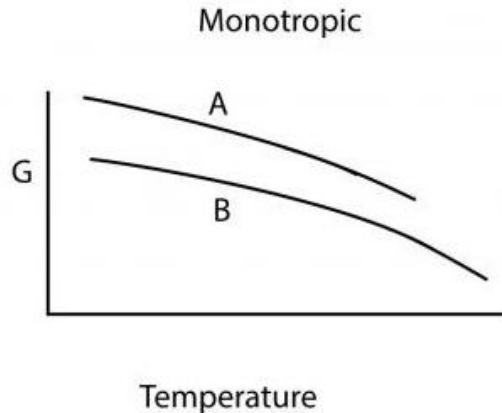
Dimorphic systems = two polymorphic forms

Monotropic polymorphic system

- Those for which only one form is stable, irrespective of temperature and pressure

Enantiotropic polymorphic system

- Different forms are stable under different external conditions



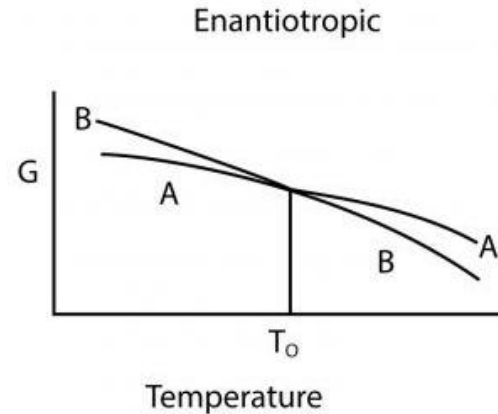
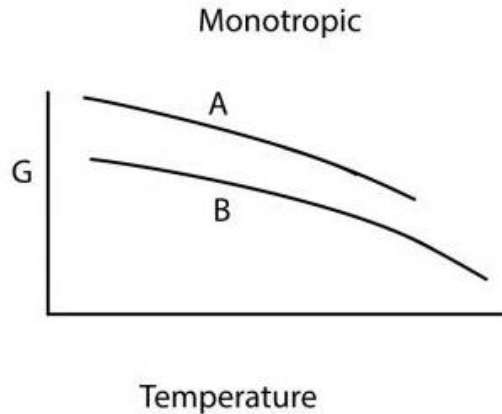
Monotropic vs. enantiotropic crystallisation

Polymorphs A and B

Monotropic system- polymorph B is more stable

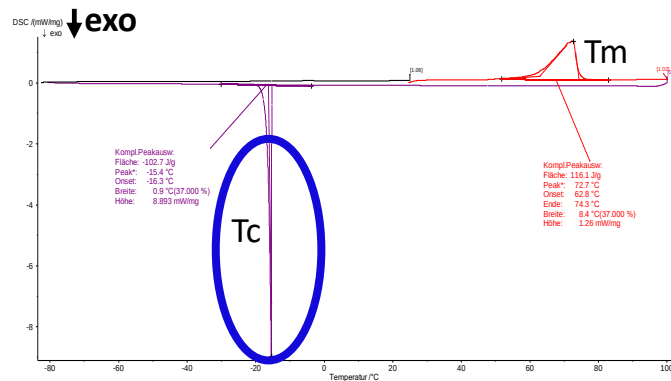
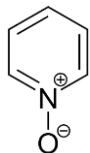
Enantiotropic system – each enantiotrope has its own T-range of stability

T_0 – transition temperature; both A and B are stable; equal free energy G

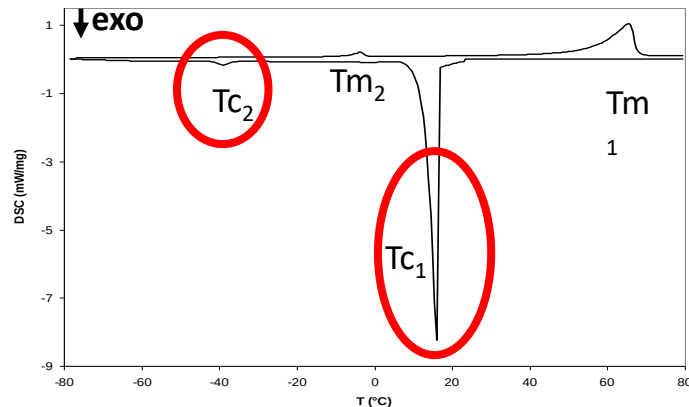


DSC: differential scanning calorimetry

➤ H/D-Pyridine N-oxide Polymorphism

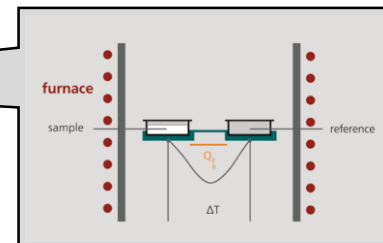


Pyridine N-oxide
Form I



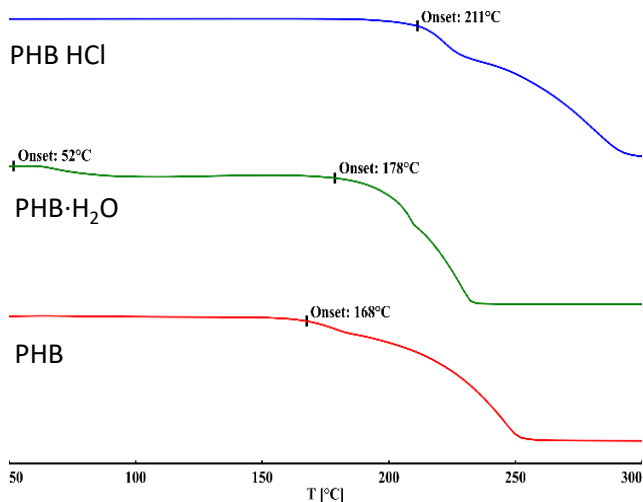
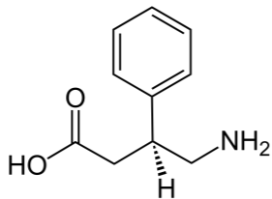
2,3,6-Trideuteropyridine N-oxide
Forms I & II

- solid-solid phase transition
- gas phase transition
- melting
- crystallisation
- decomposition



TG: thermogravimetric analysis

➤ Phenibut Hydrate



Onset temperature for decomposition

Information about:

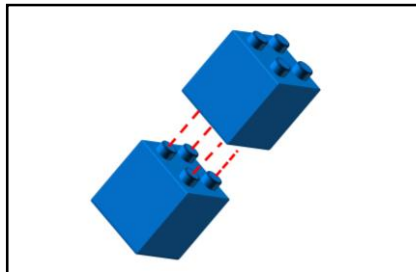
- decomposition temperatures
- decomposition steps
- correlated mass loss
- inclusions like hydrate and solvate molecules

- Different stability for polymorphs
- Check competing hydrate/solvate vs. co-crystal formation
- Thermal stability studies

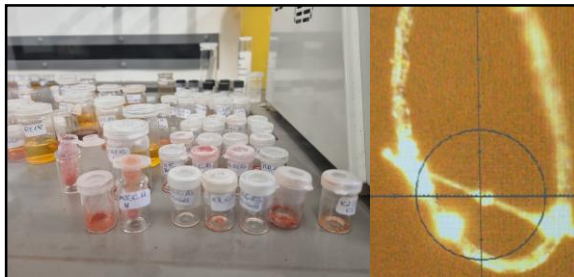
From theory & basics to real examples

case studies of pharmaceutical & optical functional materials

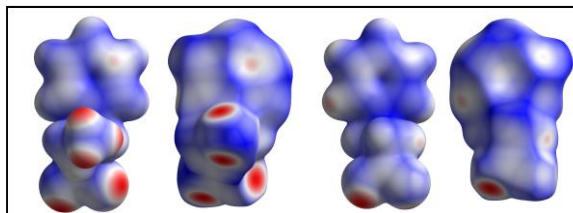
Understanding structures



Screening + analytics



Theoretical methods



Lattice energies E_{lat}

Potential application



CE with pharmaceuticals... Why???

Drug design & development → improve & fine tune **physicochemical and pharmacological properties** of drugs

therapeutic effects

- solubility
- chemical stability
- dissolution rate
- bioavailability

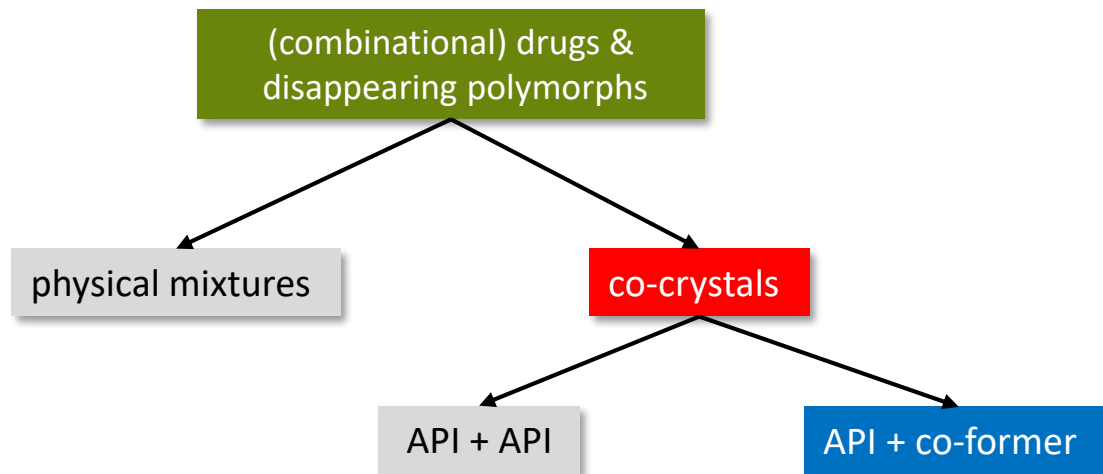
processing attributes

- crystal habit & density
- powder handling & flow
- tableting
- formulation
- ...

patentability

crystalline Active Pharmaceutical Ingredient (API)

as salts, polymorphs, solvates (incl. hydrates)

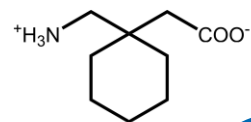


co-former limitations: GRAS & EAFUS lists
from US Food and Drug Administration (FDA)

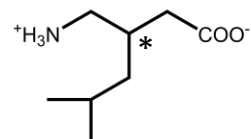
Pharmaceutical GABA-Derivatives

γ -amino butanoic acid
nonessential amino acid

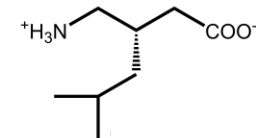
GABA



Gabapentin



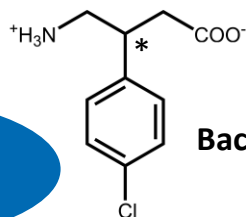
Pregabalin



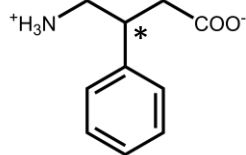
(S)-Pregabalin

enantiomer

central nervous
system
depressants



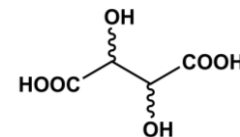
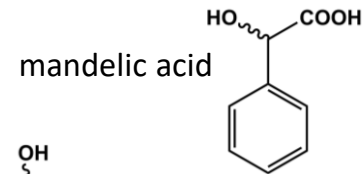
Baclofen



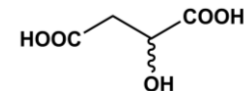
Phenibut

Similarities ?
Phase control ?
Tailored properties ?
Applications ?

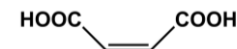
molecular complexity



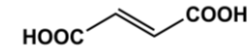
tartaric acid



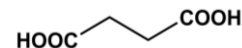
malic acid



maleic acid



fumaric acid



succinic acid

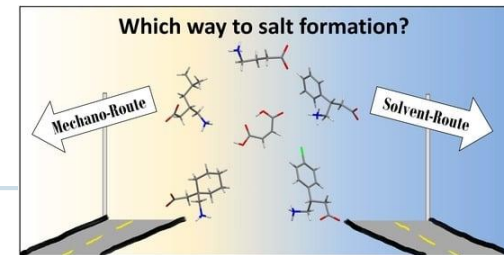
- anticonvulsant agents, pain inhibitor, anxiolytic drugs
- treatment of neuropathic pain, anxiety, insomnia, stress-related symptoms

Maleic Acid as a Co-Former for Pharmaceutically Active GABA Derivatives: Mechanochemistry or Solvent Crystallization?

Motivation & Goal

- systematic analysis of APIs with organic acids as **co-formers**
- stability studies
- increase solubility of APIs
- check the influence of different synthetic routes

- maleic acid: low cost & well-established co-former
- multicomponent crystals in a *salt-co-crystal continuum*
- crystallisation from solution & mechanochemistry



PXRD-analysis

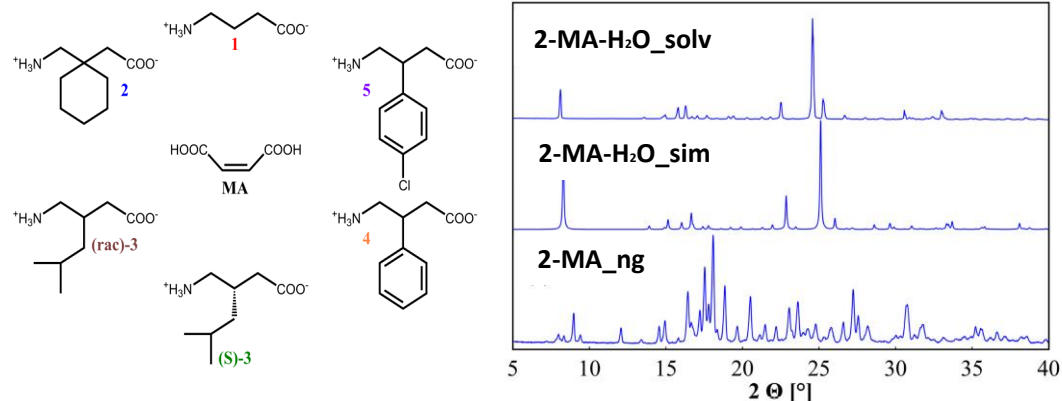


Table 1. Overview on the conducted crystallization methods for 1–5 maleates.

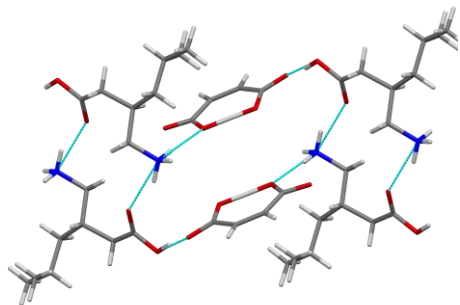
Sample	Solvent Cryst.	Neat Grind.	Liquid-Assisted Grind.
1-MA	pure phase	pure phase	not conducted
2-MA	pure 2-MA•H ₂ O	pure phase	mixture with 2-MA•H ₂ O
2-MA•H ₂ O	pure phase	pure 2-MA	mixture with 2-MA
(rac)-3-MA-I	more (rac)-3-MA-I	more (rac)-3-MA-II	not conducted
(rac)-3-MA-II	more (rac)-3-MA-I	more (rac)-3-MA-II	not conducted
(S)-3-MA•H ₂ O	pure phase	no new phase	pure phase
4-MA	impure phase	impure phase	not conducted
5-MA	pure phase	pure phase	not conducted

- impact of synthetic routes on new phase formation

Maleic Acid as a Co-Former for Pharmaceutically Active GABA Derivatives: Mechanochemistry or Solvent Crystallization?

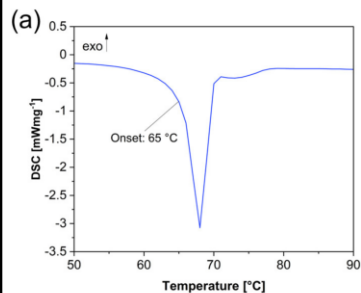
➤ characterization

(rac)-3-MA

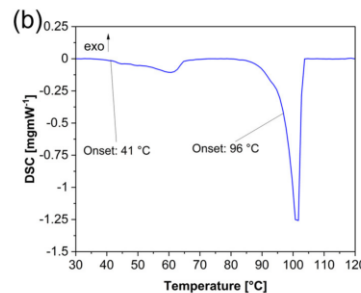


- hydrogen bonded networks
- **proton transfer**
- strong charge-assisted HBs

DSC



2-MA-H₂O solution



2-MA grinding

Sample	Melting Enthalpy [Jg ⁻¹]	Onset [°C]
1-MA	176	105
1-MA_M	178	108
2-MA•H ₂ O	135	65
2-MA_M (Peak 1)	14	41
2-MA_M (Peak 2)	85	96
(rac)-3-MA	125	98
(rac)-3-MA_M	132	98
(S)-3-MA•H ₂ O	53	46
(S)-3-MA•H ₂ O_M	87	58
4-MA	165	133
4-MA_M	181	134
5-MA	376	163
5-MA_M	362	163

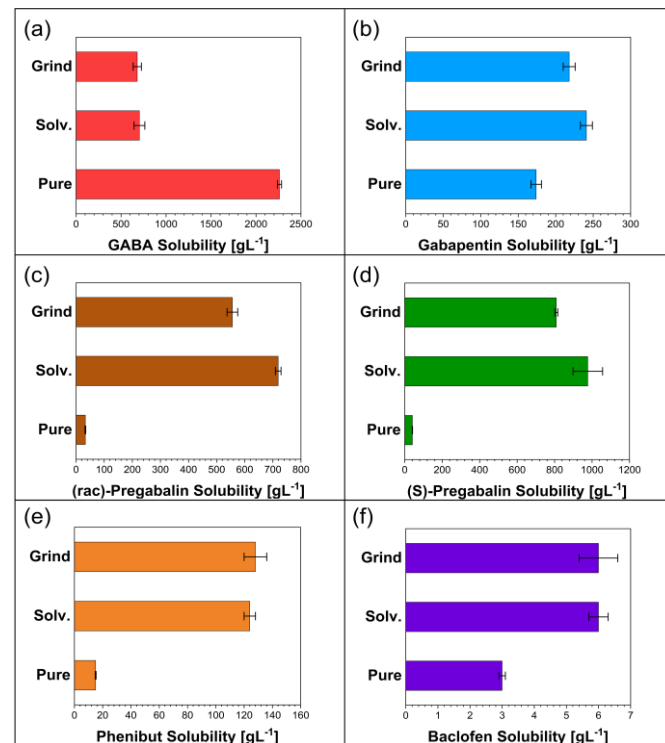
- **stabilisation of the hydrogen-bonded network**
- **thermal behaviour**
 - independent of the synthetic route
 - DSC & TGA: similar for most binary, decreased in ternary systems
 - depends on the phase formed

Maleic Acid as a Co-Former for Pharmaceutically Active GABA Derivatives: Mechanochemistry or Solvent Crystallization?

➤ solubility

	Solubility [g L ⁻¹]		Solubility [g L ⁻¹]
Maleic acid (MA)	687 ± 44		
GABA (1)	2261 ± 23	1-MA	704 ± 60
Gabapentin (2)	174 ± 7	2-MA•H ₂ O	241 ± 8
Pregabalin ((<i>rac</i>)-3•H ₂ O)	33 ± 1	(<i>rac</i>)-3-MA	719 ± 10
Pregabalin ((<i>S</i>)-3)	41 ± 1	(<i>S</i>)-3-MA•H ₂ O	977 ± 79
Phenibut (4)	15 ± 0.3	4-MA	124 ± 4
Baclofen (5)	3 ± 0.1	5-MA	6 ± 0.3

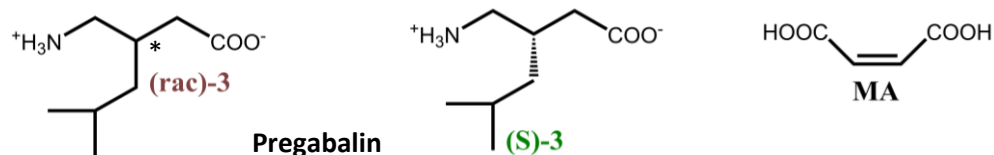
- maleates formation with all GABA-APIs
- slight differences in solubility via synthetic routes
- **solubility increase up to 23-fold except for GABA**



API solubility in a phosphate buffer at 25 °C

Maleic Acid as a Co-Former for Pharmaceutically Active GABA Derivatives: Mechanochemistry or Solvent Crystallization?

➤ processing attributes

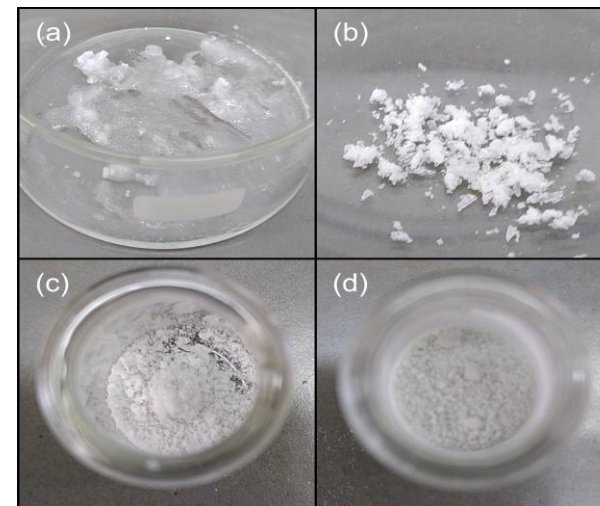


	Solvent cryst.	Grinding cryst.
Quickness	-	+
Phase reliability	=	=
Product quality	=/-	= /+
Consumption	-	+

Synthesis method can greatly affect the received phase and quality
⇒ Careful choice of the method for the desired product

(S)-3-MA • H₂O

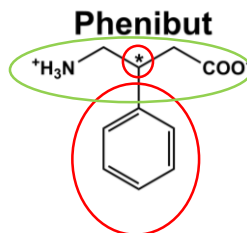
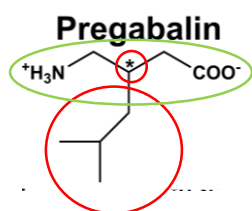
(rac)-3-MA



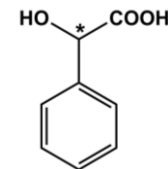
(a), (b): 3 days of drying at ambient temp. after removal from mother liquid
(c), (d): after grinding

API Adventure Ride: pregabalin & phenibut study

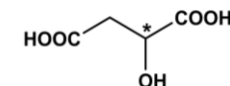
„Co-crystals of zwitterionic GABA API's pregabalin and phenibut: properties and application”



mandelic acid



malic acid



Similarities and differences:

GABA-chain

chirality

zwitterionicity

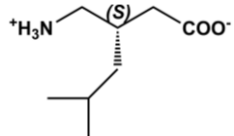
different residues

- Systematic analysis of pregabalin and phenibut with malic and mandelic acids as **co-formers**
- Where on the *salt-co-crystal continuum*?
 $\Delta pK_a = 0-3 \rightarrow$ murky area

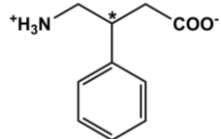
API Adventure Ride: pregabalin & phenibut study

API

(*S*)-pregabalin

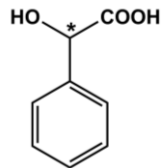


phenibut

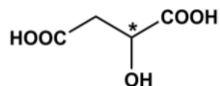


Co-former

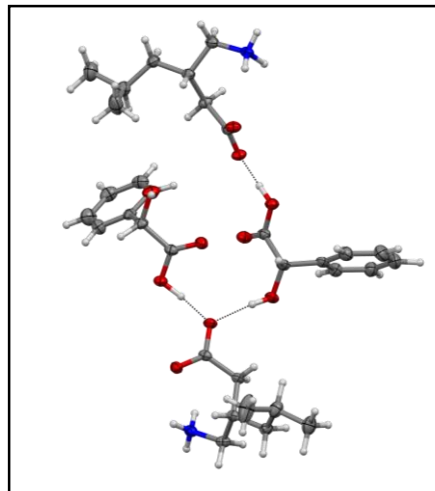
mandelic acid



malic acid

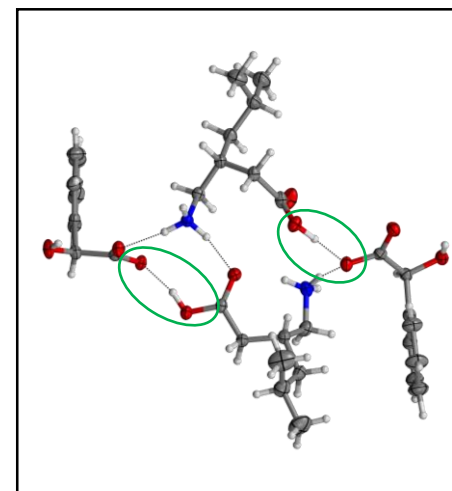


(*S*)-pregabalin:(*S*)-mandelic acid



homochiral (*S:S*) co-crystal

(*S*)-pregabalin:(*R*)-mandelic acid



heterochiral (*S:R*) salt

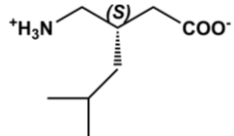
- phenibut:mandelic acid co-crystal
- **two** pregabalin:mandelic systems

- Melting points = 138 °C | 111 °C
- **Solubility:** (*S:S*) co-crystal = $37 \pm 1 \text{ g L}^{-1}$
(*S:R*) salt = **316** $\pm 18 \text{ g L}^{-1}$
(*S*)-pregabalin = $35 \pm 1 \text{ g L}^{-1}$

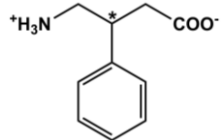
API Adventure Ride: pregabalin & phenibut study

API

(S)-pregabalin

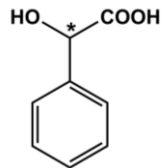


phenibut

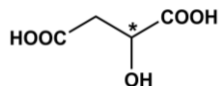


Co-former

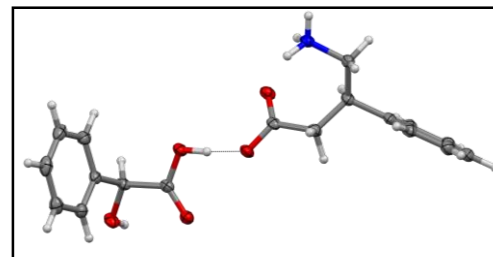
mandelic acid



malic acid



(R)-phenibut:(S)-mandelic acid



heterochiral (*R:S*) co-crystal

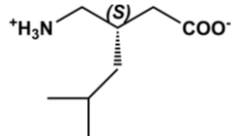
- Solubility = $71 \pm 3 \text{ g L}^{-1}$
- Melting point = 150°C
- Strong O-H $\cdots$ O hydrogen bond
- Formation seems unfavoured

➤ phenibut:mandelic acid co-crystal

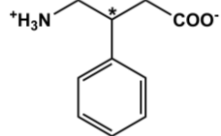
API Adventure Ride: pregabalin & phenibut study

API

(S)-pregabalin

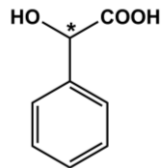


phenibut

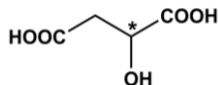


Co-former

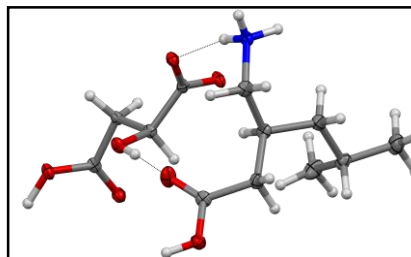
mandelic acid



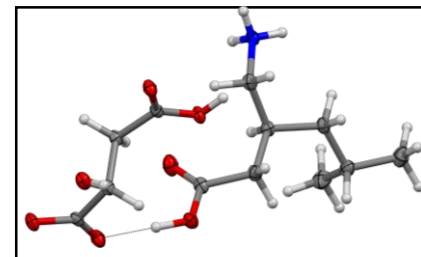
malic acid



(S)-pregabalin:(S)-malic acid



(S)-pregabalin:(R)-malic acid



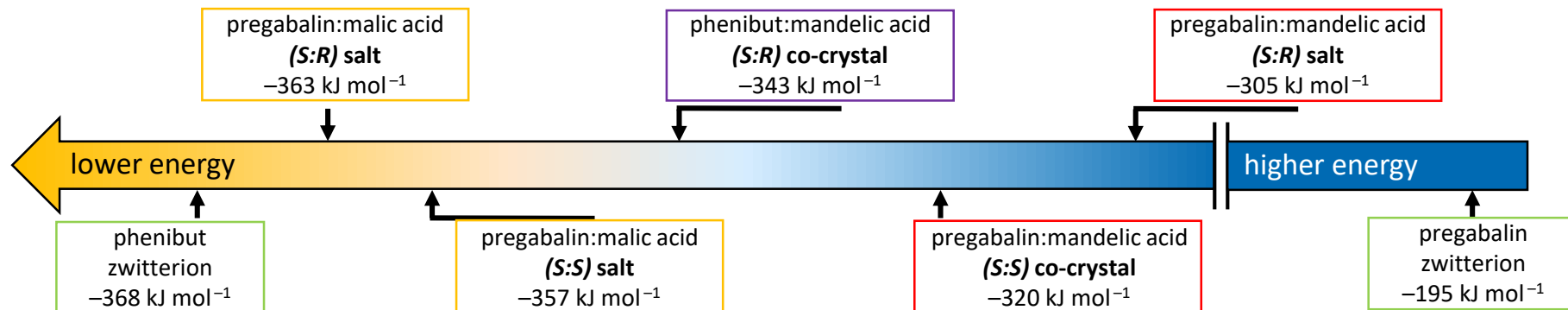
homochiral (**S:S**) & heterochiral (**S:R**)
salts

- Solubility = **>800** g L⁻¹
- Melting point = 85 °C | 95 °C
- Similar crystal structures
- Strong O-H...O hydrogen bonds

- two pregabalin:malic systems
- **NO** phenibut:malic acid system

Why ???

Lattice energies (E_{lat}) landscape



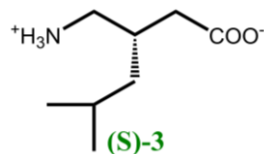
Formation of multicomponent systems = energy win for the system

The lower E_{lat} = the higher the stability

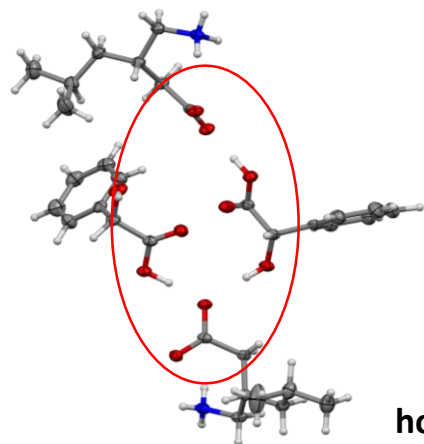
➤ **Phenibut is the most stable & hardly undergoes co-crystallisation**

Pregabalin:mandelic acid – salt vs. co-crystal

(S)-Pregabalin



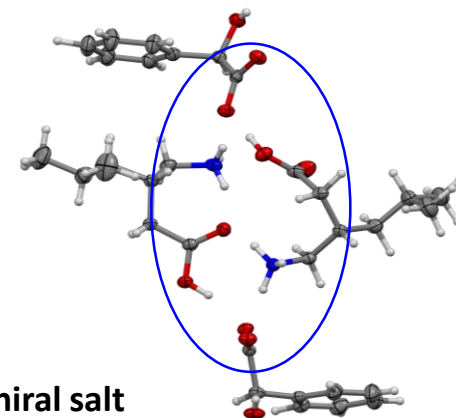
(S)-mandelic acid



homochiral co-crystal

no proton transfer

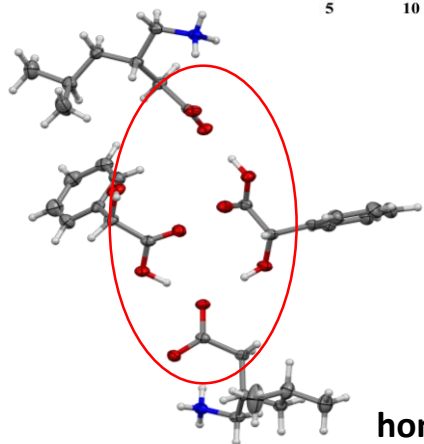
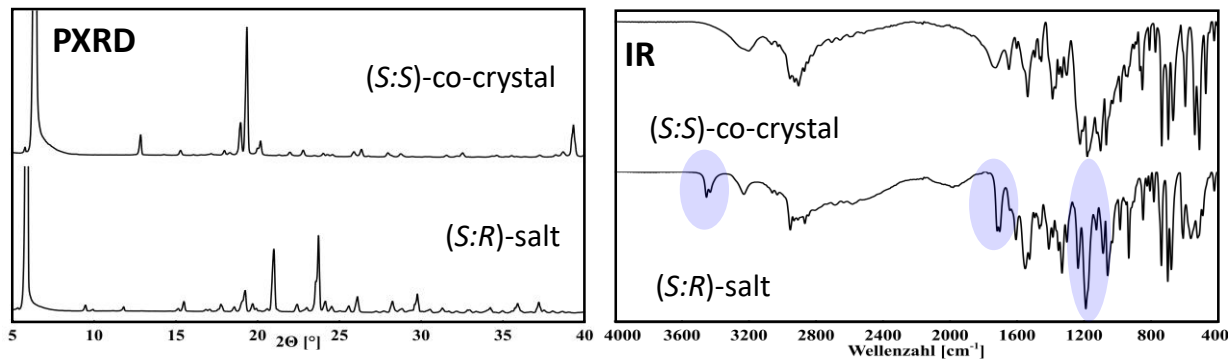
(R)-mandelic acid



heterochiral salt

deprotonation of mandelic acid

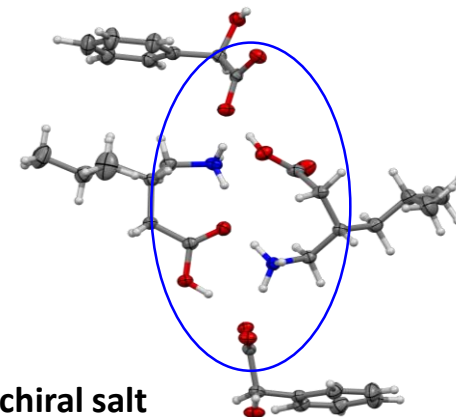
Pregabalin:mandelic acid – salt vs. co-crystal



homochiral co-crystal

no proton transfer

- Different powder patterns
 >> two distinct solid phases
- Protonation status confirmed by IR
 >> $\nu(\text{s})$ O-H
 >> changes in $\nu(\text{s})$ C=O and $\nu(\text{s})$ C-O



heterochiral salt

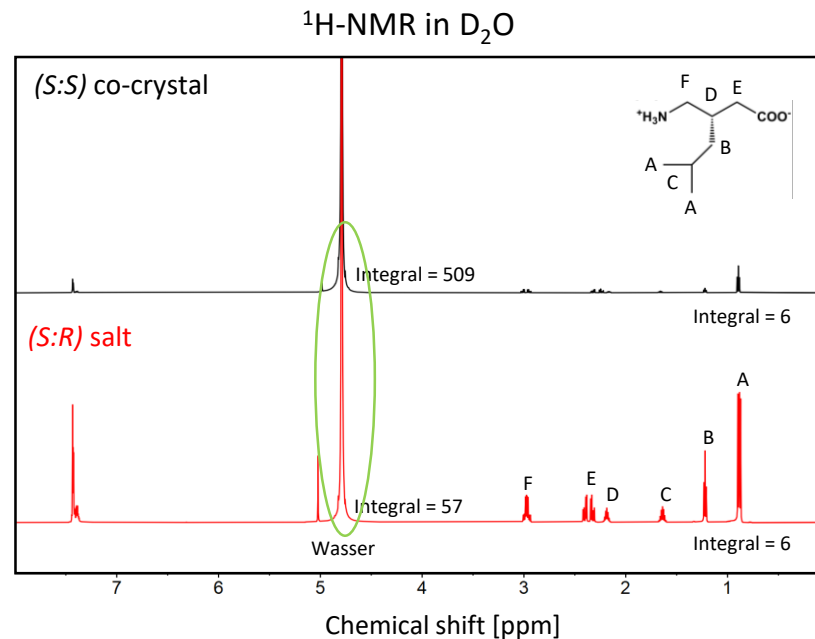
deprotonation of mandelic acid

Solubility of pregabalin:mandelic acid systems

	Solubility [g L ⁻¹]	Mp [°C]	E _{Lat} [kJ mol ⁻¹]
(S)-Pregabalin	35 ± 1	185	-195
(S:S)-co-crystal	37 ± 1	138	-320
(S:R)-salt	316 ± 18	111	-304

Solubility determination with ¹H-NMR

- >> comparison to water signal
- >> multiple determinations
- >> simple but errors up to 5-10%

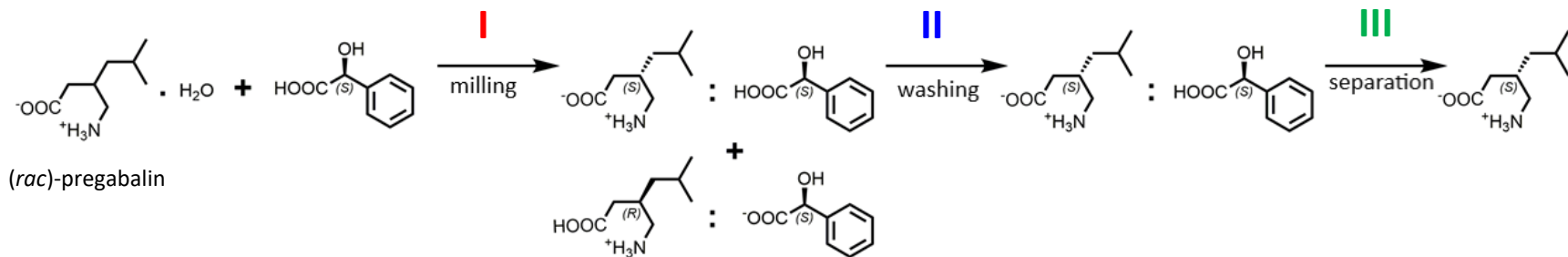


Significant differences in melting points and solubilities of products!
What can we do with this knowledge?

Toward application: racemate separation

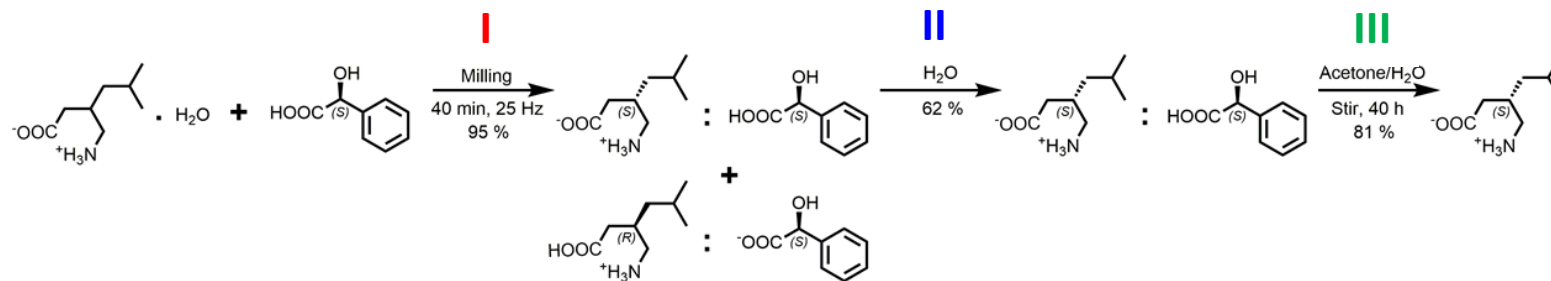
(*S*)-Pregabalin – **eutomer** = enantiomer with **desired pharmaceutical properties**

Aim: easy racemate separation via co-crystallization
use significant differences in solubility
pregabalin:mandelic acid (*R:S* / *S:R*) salt and (*S:S* / *R:R*) co-crystal

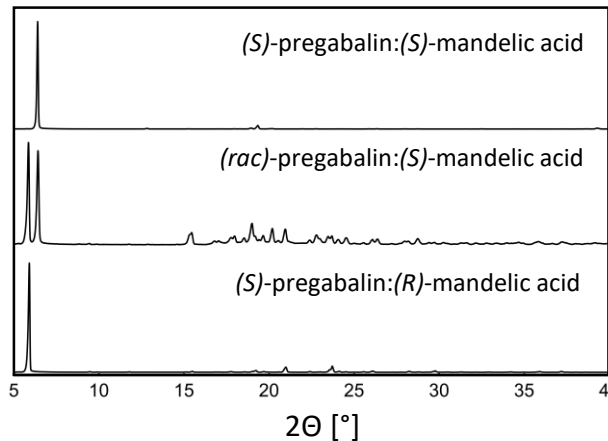


- I: „green“ mechanochemical synthesis of homochiral and heterochiral mixtures in a short time
- II: Only very small amounts of water required
- III: Recycling of acetone & mandelic acid $\rightarrow$ extraction without heating

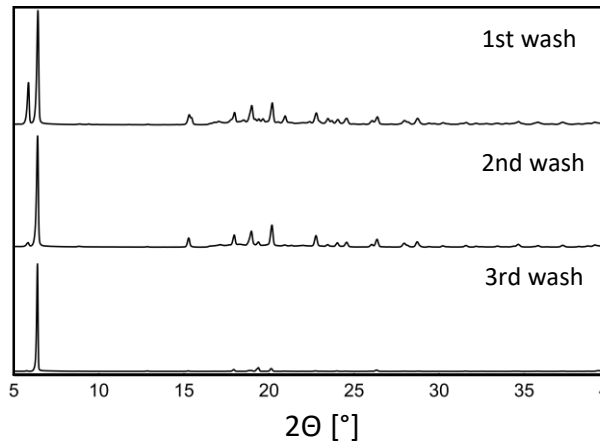
Think it's pure? Think again!



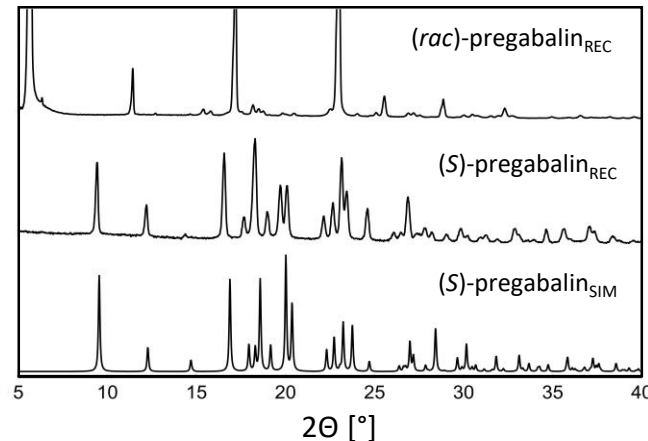
Step I



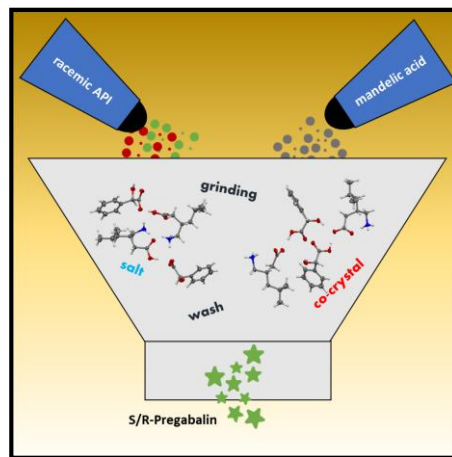
Step II



Step III



Toward application: optimised top-down pregabalin enantiopurification



Aim

racemate separation
via co-crystallization
based on different
solubilities

- non-destructive method
- resources recyclable
- unproblematic substances
- low solvent consumption
- no heating
- control with PXRD

Outcome

- ✓ improved and simplified
 - ✓ more sustainable ("green")
 - ✓ low-effort
 - ✓ easily controllable
- top-down enantiopurification
process**

Towards new functional optical materials

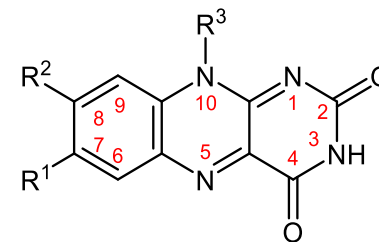
- understand and modulate optical properties
- implement intermolecular interactions
- multi-component systems as an alternative to classical route
- Potential application in OLEDs

lack of solid-state research

co-crystals

binary & ternary
systems

no limitation for
co-former



isoalloxazine: $R^1 = R^2 = H$

flavin group: $R^1 = R^2 = CH_3$

Flavins

- chromophores with a central role in biological processes (riboflavin - vitamin B₂)
- electronic structure & optical spectra are sensitive to settings*
- are considered
 - sustainable
 - structurally versatile
 - easily synthesizable

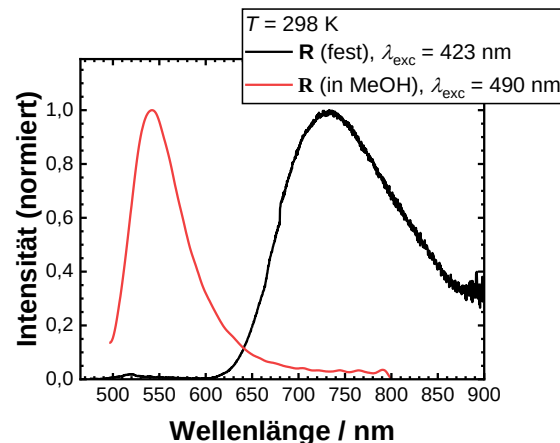
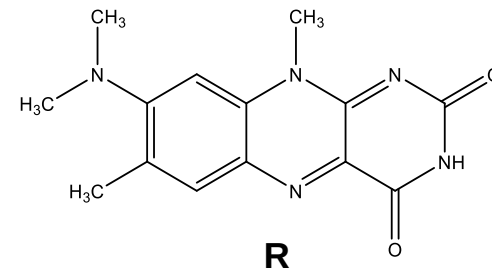
Model flavin: roseolumiflavin

- relatively small, chemically stable, rigid & possesses multiple functional groups suitable for rational design strategies

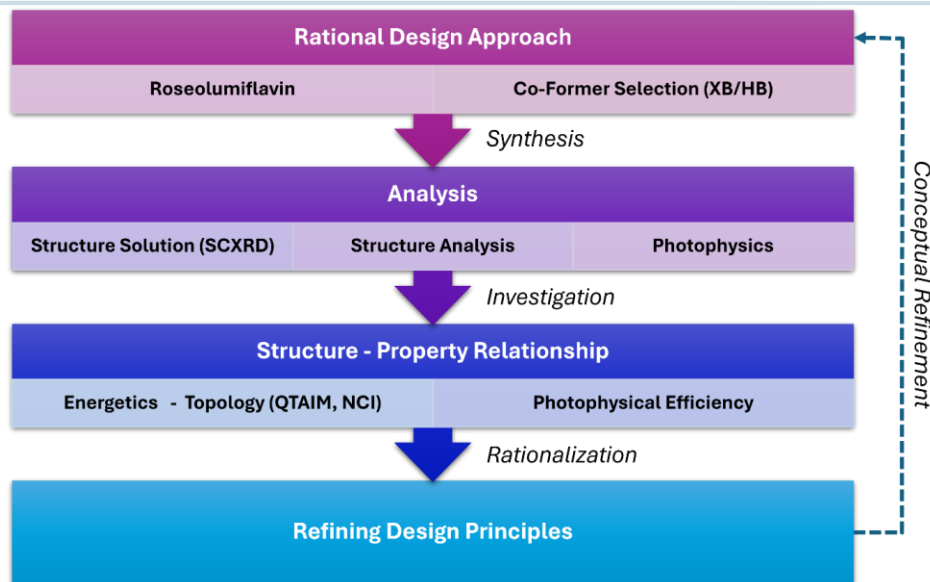
Properties

- deep red colour in the solid state
 - reported no luminescence - ACQ (aggregation-caused quenching)?
 - emission in a near-IR region – no ACQ
- pronounced solvatochromism in solution
- possible directional intermolecular interactions
- electron density delocalized across the ring framework

Goal: Systematic investigation of structure-property relationships between the supramolecular structure and the emission behavior of flavins on a model compound



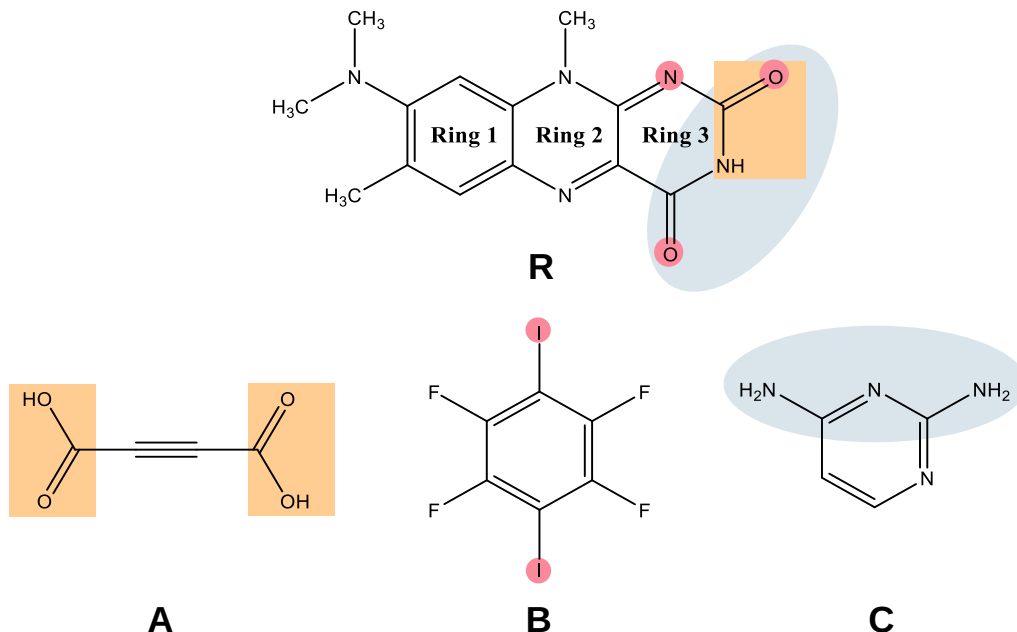
Workflow strategy



- i. Identify robust pathways to utilize the flavin's imide group to develop MCC and test the possibility of actualization of different supramolecular motives (involving HB and XB)
- ii. Structurally determine and compare the resulting packing motifs
- iii. Correlate structural differences with solid-state photophysical properties
- iv. Rationalize packing preferences and interaction hierarchies using energetical approach and topological analyses
- v. Establish mechanochemical routes as a sustainable synthetic approach for the selective crystallization of multicomponent systems.

Ansatz: Modell-Co-Former als funktionale Bausteine für HB- oder XB- verknüpfte supramolekulare Synthese

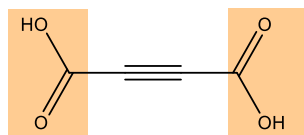
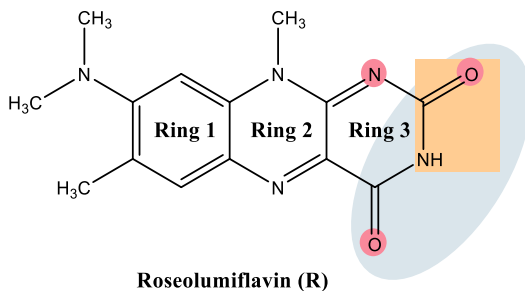
- **A** – Acetylendicarbonsäure
 - Zwei Carboxygruppen
 - Lineares Molekül
- **B** – 1,4-Diiodotetrafluorbenzol
 - Halogengruppendonor
 - Aromatischer Ring
- **C** – 2,4-Diaminopyrimidin
 - Donor-Akzeptor-Donor (DAD)
 - HB-Anknüpfstelle
 - Aromatischer Ring



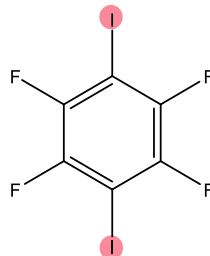
Proof-of concept: co-crystallisation of roseolumiflavin

➤ find suitable co-former

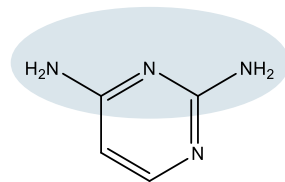
- utilize imide group: donor & acceptor functionalities
- generate one-, two- and three-point recognition patterns
- intermolecular interactions
 - hydrogen bonds
 - halogen bonds
 - stacking interaction
- solid-state emission?



A



B



C

A: acetylene dicarboxylic acid

- two carboxyl groups
- linear, rigid molecule

B: 1,4-Diiodotetrafluorobenzene

- halogen group donor
- aromatic ring

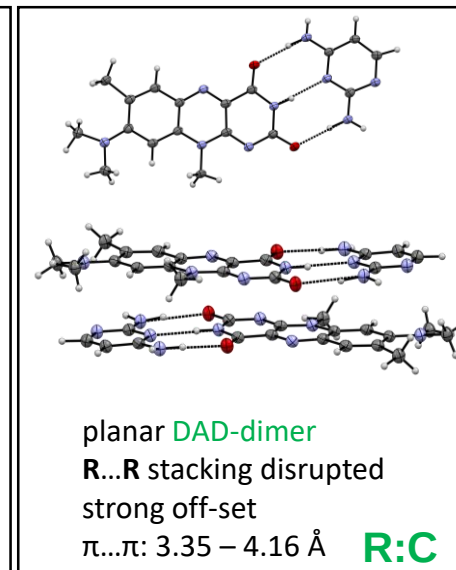
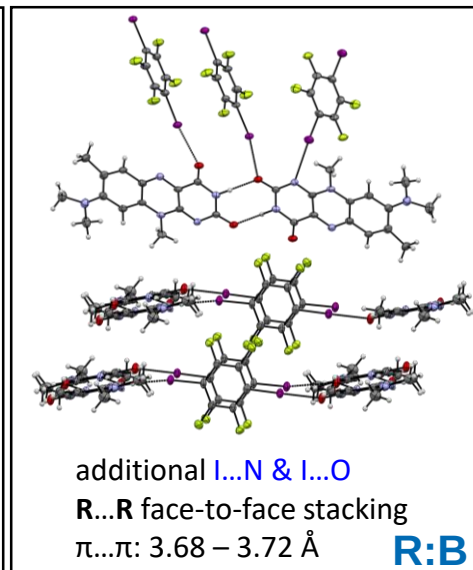
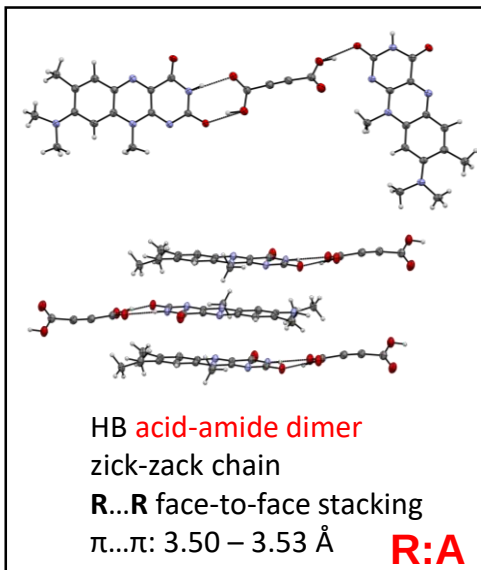
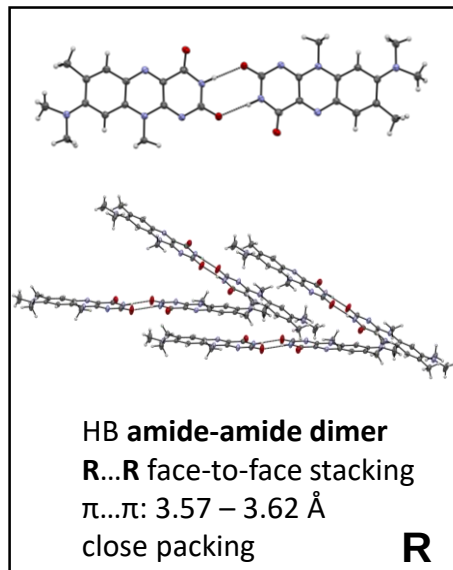
C: 2,4-Diaminopyrimidine

- donor-acceptor-donor (DAD)
- HB binding site
- aromatic ring

Proof-of concept: co-crystallisation of roseolumiflavin

➤ structural outcome

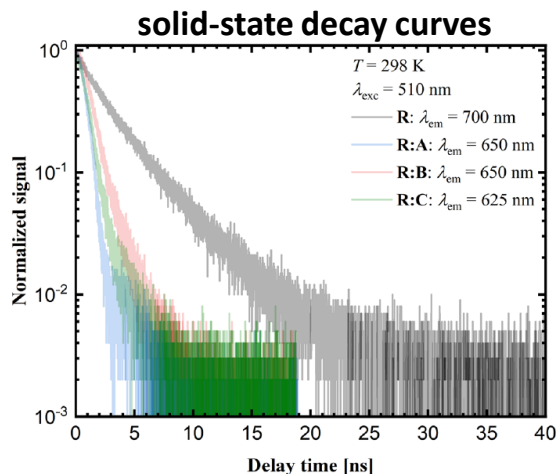
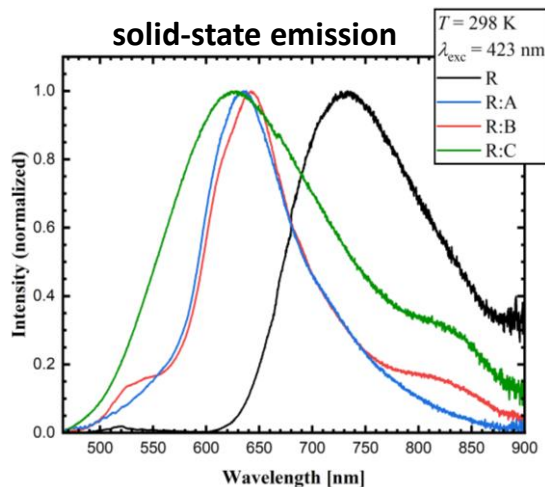
- concept proved
- one-, two- and three-point recognition patterns
- intermolecular interactions - **hydrogen bonds** - **halogen bonds** - **stacking interaction**



Proof-of concept: co-crystallisation of roseolumiflavin

➤ optical properties

- **blue shift** upon co-crystallisation
- small Stokes shifts
- excitation from the ground into the locally excited state & charge-transfer effects
- **AIE** (aggregation-induces emission) in **R**



	R	R:A	R:B	R:C
daylight				
UV-lamp 366 nm				
$\lambda_{\text{em}} [\text{nm}]$	733	638	643	628
$\tau [\text{ns}]$	2.85	0.99	1.42	0.98

Correlation between emission and degree of the crystal packing alteration:

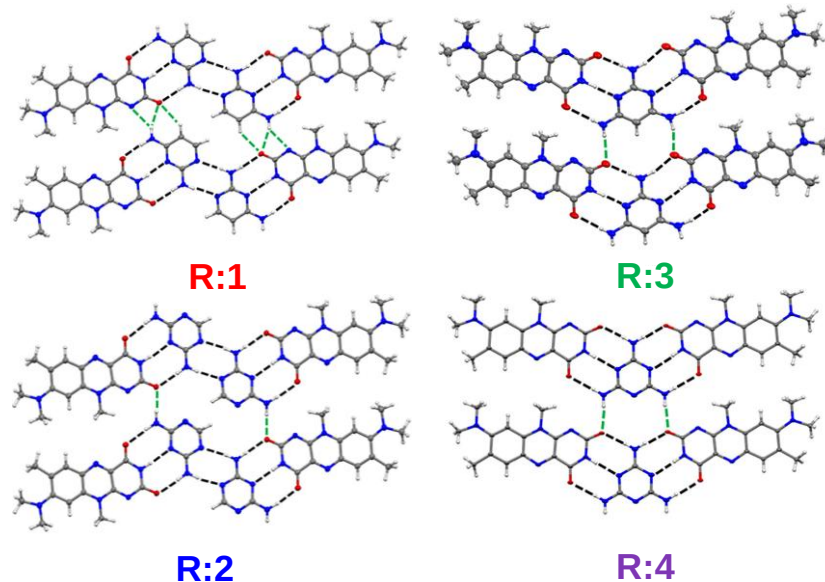
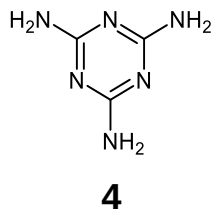
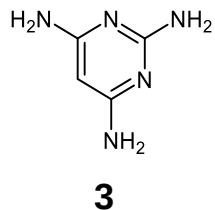
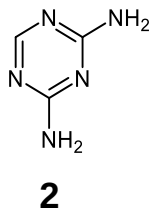
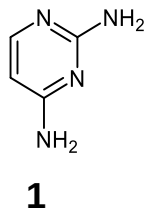
- general alteration of the packing has higher impact than particular co-crystal

Next step: fine tuning the co-formers

Approach:

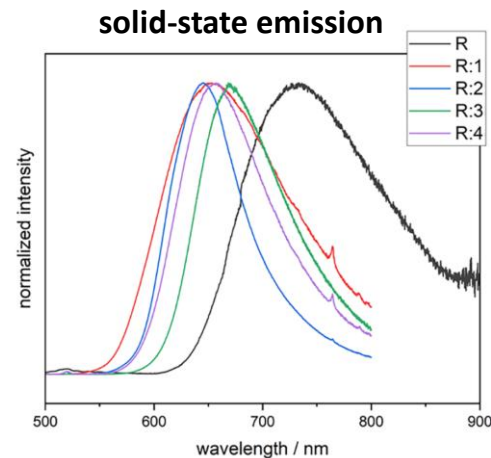
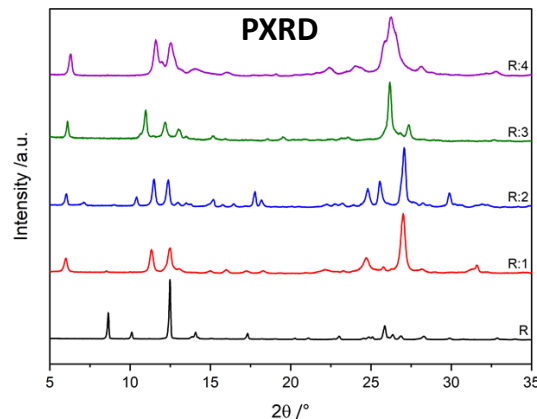
- systematic co-former variation based on *N*-heterocycles
- variation of basicity, geometry, HB donor/acceptor ability
- **binary MCC**

Effect of the gradual modification of the supramolecular environment ?



- three-point-recognition pattern
- **R:1** and **R:2**: tetrameric structural motif
- **R:3** and **R:4**: trimeric structural motif
- chain linkage lateral to the structural motif

- new **binary** co-crystals with all *N*-heterocycles as co-formers
- phases reproducible
- eco-friendly mechanochemistry
- photophysics:
 - solid-state emission **blue shifted**
 - **narrowed** emission band
 - smaller Stokes shifts



	$\lambda_{\text{em,max}}$ (nm)	$\lambda_{\text{ex,max}}$ (nm)	FWHM (Gaussian fit)	Stokes shift (nm)	Stokes shift (cm ⁻¹)	CIE 1931 coordinates (x, y)
R	733	633	158	100	2155	0.64, 0.35
R:3	668	622	97	46	1107	0.70, 0.30
R:4	657	582	102	75	1961	0.68, 0.31
R:1	652	613	141	39	976	0.65, 0.35
R:2	646	583	80	63	1673	0.68, 0.32

From binary to ternary systems

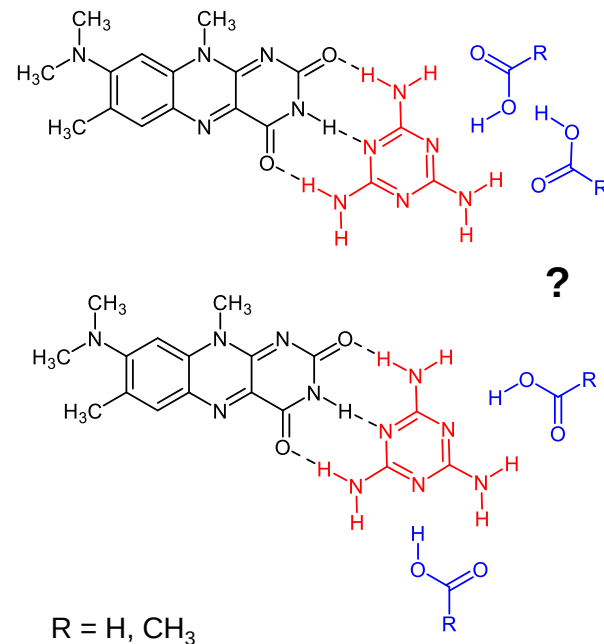
- include a third component

Carboxylic acids: formic acid (F) and acetic acid (A)

Hypothesis:

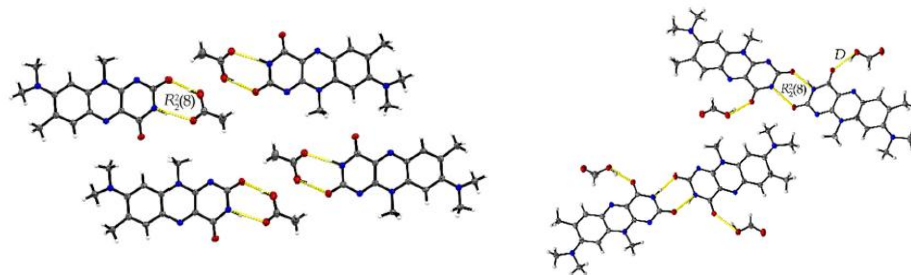
- increased structural complexity for more precise control of packing
- additional HB donors/acceptors generate competing motifs
 - further alteration of crystal packing
 - fine-tuning of solid-state photophysics

Why carboxylic acids?

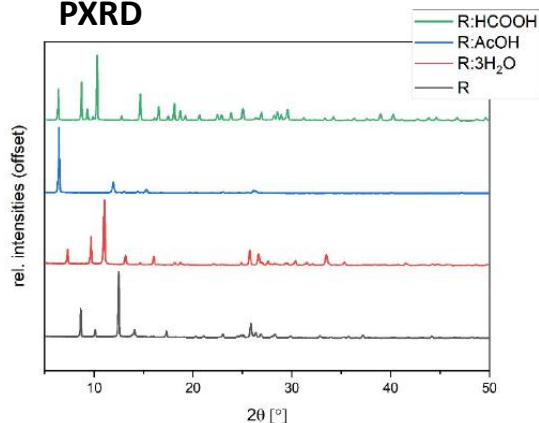


Stability of roseolumiflavin solvates

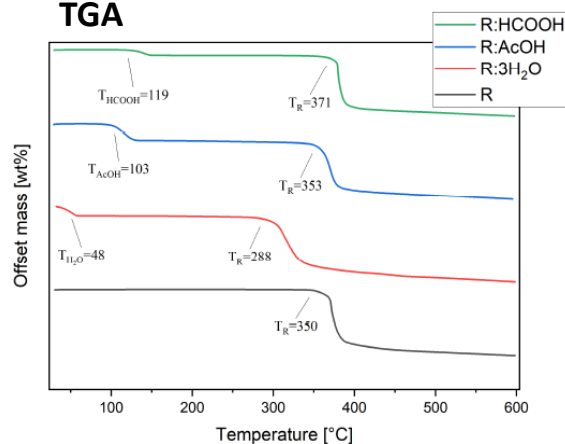
- R - well soluble in just few organic solvents
- but high solubility in formic and acetic acids
- stabilisation upon hydrate & solvate formation
- no proton transfer



PXRD



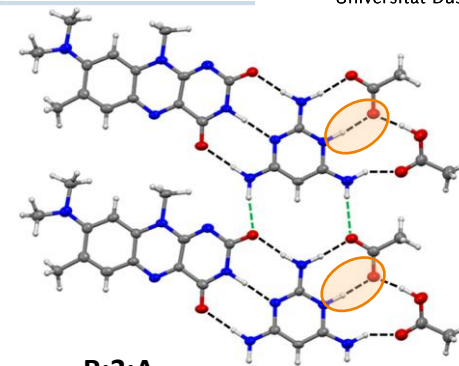
TGA



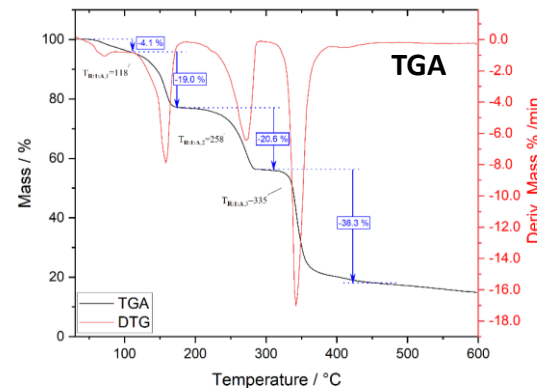
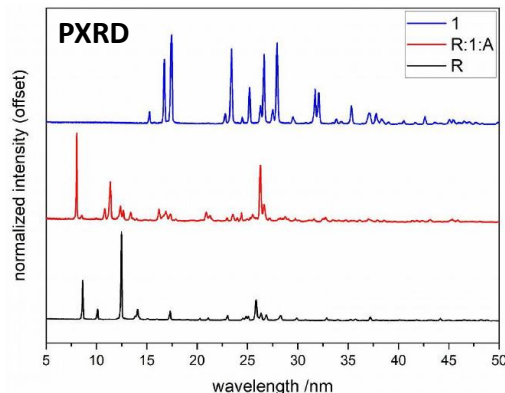
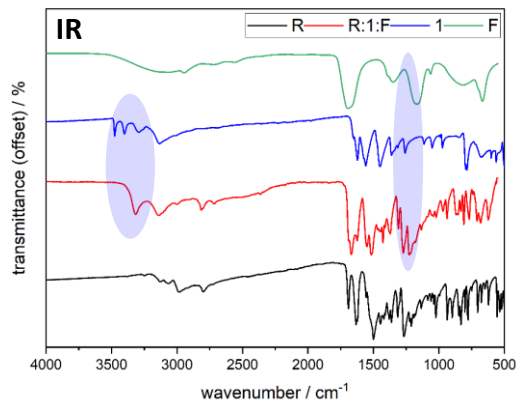
Structure	Quantum Espresso
	E_{lat} [kJ/mol]
R	-181.06
R:HCOOH	-197.58
R:AcOH	-283.55
R:3H ₂ O	-395.82

Idea:
can we introduce
carboxylic acids in
ternary MCCs?

- introduced additional third component – 6 **ternary** systems
- hydrogen bonded **DAD** and **pseudo DAD-motive**
- reproducible: mechanochemistry and/or solvent
- Protonation on co-former
 - partial charge-transfer
 - contain both salt-like and cocrystal-like features
 - molecular ionic cocrystals

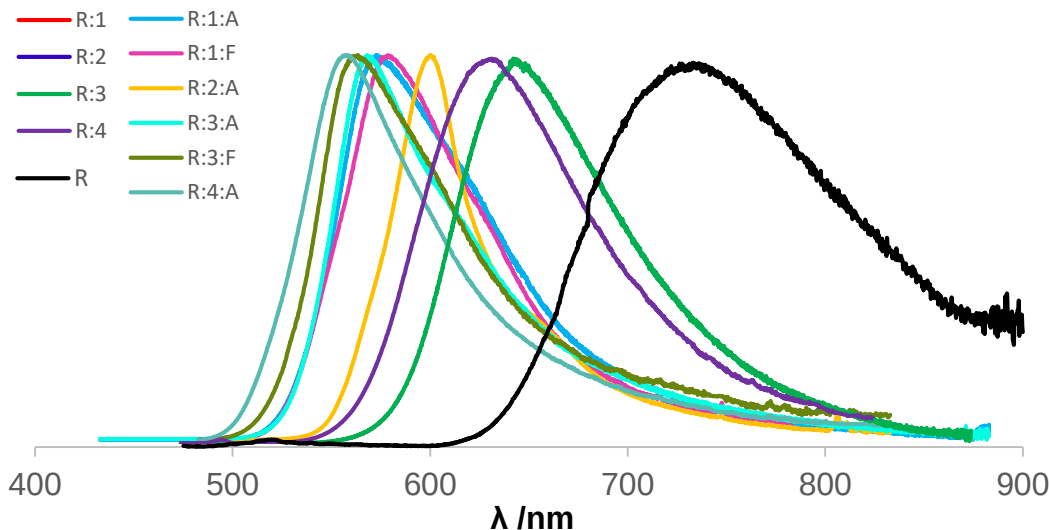


R:3:A



Solid-State Emission in R binary & ternary MCCs

- solid-state emission even more **blue shifted**
- mixed ionized systems emit even **narrower bands**
- not all parameters are consistently decreasing or increasing
→ **possibility to chose the combination according to the problem**

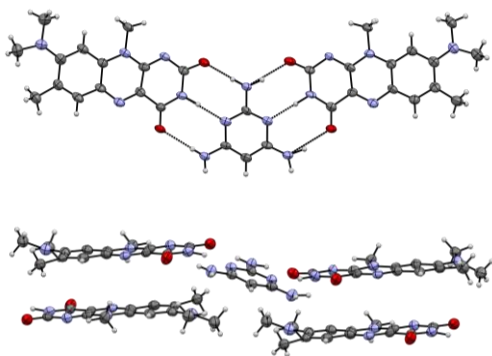


	$\lambda_{\text{Em,max}}$ (nm)	$\lambda_{\text{Ex,max}}$ (nm)	FWHM (Gaussian fit)	Stokes shift (nm)	Stokes shift (cm ⁻¹)
R	733	633	158	100	2155
R:3	668	622	97	46	1107
R:4	657	582	102	75	1961
R:1	652	613	141	39	976
R:2	646	583	80	63	1673
R:2:A	617	575	50	42	1184
R:1:F	596	542	85	54	1570
R:1:A	590	545	89	45	1399
R:3:A	585	556	76	29	892
R:3:F	581	560	81	21	654
R:4:A	575	539	78	36	1162

Understand structure-property-relation

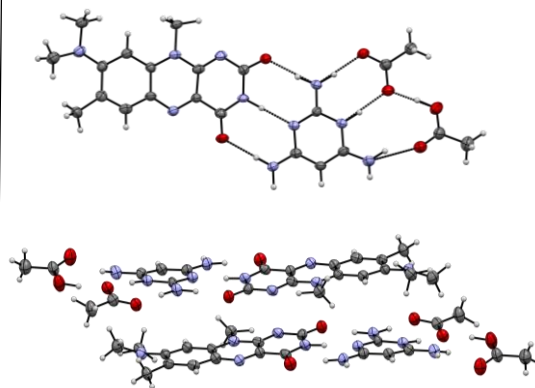
with Non-Covalent Interactions NCI-plots

R:4



- non-planar DAD-trimer
- R...R stacking
- face-to-face
- $\pi\cdots\pi$: 3.29 – 3.47 Å

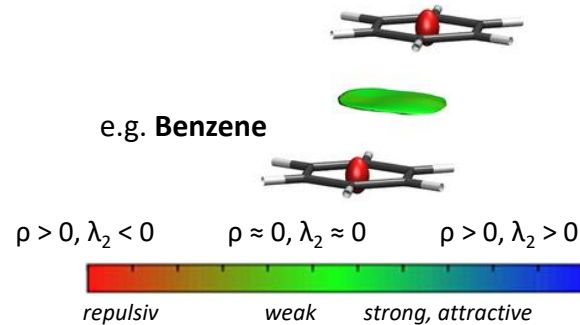
R:4:A



- planar DAD-tetramer
- proton transfer
- R...R stacking disrupted
- strong off-set
- $\pi\cdots\pi$: 3.26 – 3.65 Å

- identification and visualization
- analysis of regions with low electron density gradient
- qualitative characterization

e.g. Benzene



Modified superposition motifs:

R: parallel offset, charge-transfer character

binary: nearly face-to-face, altered NCI contact areas

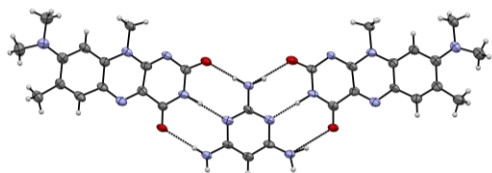
ternary: co-former rings - larger contact area

Understand structure-property-relation with energies of intermolecular interactions

- **calculated interaction energy** for all HB
- quantitative characterization
 - >> charge-assisted HB are much stronger (up to -70...-80 kJ/mol)
 - >> neutral HB ca. -20...-35 kJ/mol
- identify BSM (**basic structural motive**) – the most strongly bonded

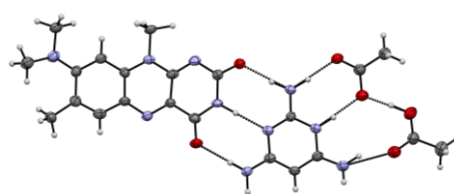
	BE R...Y (kJ/mol)	BE BSM (kJ/mol)
R	-39.00	-39.00
R:1	-65.40	-165.81
R:2	-56.48	-150.67
R:3	-69.75	-134.31
R:4	-52.68	-105.35
R:3:A	-83.22	-260.38
R:3:F	-57.45	-225.89
R:4:A	-59.50	-239.53

R:4



non-planar DAD-trimer
= BSM

R:4:A



planar DAD-tetramer = BSM

- charged-assisted HB

Summary

1. Main driving force – strong HB motives - energies
2. Secondary – stacking – NCI plots
 - >> alteration of the packing
 - >> changes of photophysical properties

Outlook: transferability to more complex flavins